Global Determinants of Life Expectancy: A Multivariate Analysis Using an ELT Data Pipeline

Data Operations & Management
CA3: Data Project

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I. Abstract:

Life Expectancy is a key measure of population health and socio-economic development, reflecting the combined effects of economic conditions, environmental factors, healthcare systems, and public health interventions. This project analyses global life expectancy movements from 2002 to 2022 by integrating multiple datasets sourced from Our World in Data website, including Life Expectancy, Gross Domestic Product (GDP) per capita, Carbon Emissions, Healthcare spending per person, Vaccination Coverage, Child Mortality, and Diet Compositions. A complete data pipeline was developed to extract, merge, and transform different datasets into a unified analytical data frame. Exploratory data analysis and visualisation techniques were applied using Tableau to examine relationships between life expectancy and its potential drivers. The results indicate a substantial global increase in life expectancy over the study period, with strong positive associations observed between GPD per capita, vaccination coverage, and life expectancy, alongside with a clear inverse relationship between vaccination rates and child mortality. However, higher healthcare spending per capita does not consistently translate into proportionally higher life expectancy, indicating differences in healthcare system efficiency. In addition, economic development is associated with increased carbon emissions, raising concerns regarding sustainability. These findings highlight the importance of targeted vaccination programmes, efficient healthcare investment, and sustainable economic growth in improving global health outcomes, while also emphasising key data governance considerations related to data quality, transparency, and ethical use of aggregated health data.

II. Introduction:

1. Background & Motivation

Life expectancy is widely recognised as one of the most important indicators of a nation's overall health and development. It encapsulates the cumulative effects of economic prosperity, access to healthcare, public health interventions, environmental quality, and nutrition. Over the past century, global life expectancy has increased significantly due to economic development, medical advances, improved sanitation, and widespread vaccination programmes. However, substantial disparities remain between high-income and low-income countries, highlighting persistent global inequalities in health outcomes. Understanding the factors that contribute to these differences is essential for informing effective public policy and international development strategies.

Vaccination coverage, healthcare spending, economic growth, environmental sustainability, and dietary patterns are frequently cited as major determinants of population health. For instance, childhood immunisation programmes have been shown to dramatically reduce preventable diseases and child mortality, while economic growth enables governments to invest in healthcare infrastructure. Conversely, rapid industrialisation often leads to increased environmental pollution, which may negatively affect long-term health outcomes. These complex and interconnected factors motivate the need for a data-driven analysis that examines life expectancy from a comprehensive global perspective.

2. Problem Statement

The objective of this project is to investigate how life expectancy varies across

countries and over time, and to identify the socio-economic, healthcare, environmental, and nutritional factors most strongly associated with these variations. Specifically, the project addresses the following research questions:

- How has global life expectancy changed from 2002 to 2022?
- How do key socio-economic and health factors - including GDP, healthcare spending, vaccination coverage, emissions, and diet - influence global life expectancy over time?
- What actionable insights can be derived to inform policy and industry decision-making?

3. Mini Literature Review

Life expectancy has long been recognised as a comprehensive indicator of population health and socio-economic development, influenced by a complex interaction of economic, healthcare, environmental, and lifestyle factors. A substantial body of literature has examined the determinants of life expectancy across countries, highlighting the importance of income, public health interventions, healthcare expenditure, environmental sustainability, and nutrition.

Economic development, commonly measured using gross domestic product (GDP) per capita, is consistently identified as a key driver of improvements in life expectancy. Cross-country studies indicate a strong positive relationship between GDP and life expectancy, particularly in low- and middle-income countries, where increases in income enable improved access to healthcare, sanitation, education, and nutrition (Zaman et al.). However, the marginal gains in life expectancy diminish at higher income levels, suggesting that economic growth alone is insufficient to guarantee continuous health improvements. Jabaa et al. further demonstrate that while income growth contributes to longer life expectancy, its effectiveness depends on how resources are allocated within healthcare systems.

Healthcare expenditure has also been widely studied as a determinant of life expectancy, though findings suggest that higher spending does not always result in proportionally better outcomes. Jabaa et al. show that although health expenditure positively affects life expectancy, the strength of this relationship varies significantly across countries and over time. Similarly, Zaman et al. find that GDP per capita often exhibits a stronger association with life expectancy than total health expenditure, indicating that system efficiency, governance, and preventive care play a crucial role. These findings are particularly relevant to high-income countries, where healthcare spending is substantial but health outcomes may lag behind peer nations.

Vaccination coverage represents one of the most effective and cost-efficient public health interventions, especially in reducing child mortality. The World Health Organization highlights the role of the Global Alliance for Vaccines and Immunization (GAVI) in expanding access to essential childhood vaccines, including diphtheria–tetanus–pertussis (DTP), measles, and polio, leading to significant reductions in preventable deaths (Bustreo et al.). The literature consistently reports a strong inverse relationship between vaccination rates and child mortality, particularly in low-income settings. These findings underscore the importance of immunisation programmes in improving population health and increasing life expectancy, especially during early childhood.

Environmental factors have received increasing attention in life expectancy research, particularly in the context of economic growth and industrialisation. While higher GDP is associated with improved health outcomes, it is often accompanied by increased carbon emissions and environmental degradation. Rjoub et al., using time-series and causality analysis for Turkey, demonstrate that economic growth contributes to higher emissions, which in turn exert a negative effect on life expectancy in the long run. Although their study is country-specific, it provides causal evidence supporting concerns raised in broader cross-country analyses. This highlights a critical policy challenge: balancing economic development with environmental sustainability to ensure long-term health gains.

Dietary patterns and nutrition also play a significant role in determining life expectancy, although their effects are often more difficult to quantify using aggregated data. Fadnes et al. estimate the impact of food choices on life expectancy and conclude that dietary quality, particularly the balance of plant-based and animal-based foods, has a meaningful influence on long-term health outcomes. Their findings suggest that improvements in diet composition may lead to measurable gains in life expectancy, beyond increases in total calorie intake alone. This aligns with broader public health literature emphasising the role of nutrition in preventing chronic diseases, though detailed dietary data are often unavailable at the global level.

Overall, the existing literature supports the inclusion of economic, healthcare, environmental, vaccination, and nutritional variables when analysing life expectancy. However, many studies focus on individual determinants in isolation or are limited to specific countries or regions. This project contributes to the literature by integrating multiple global datasets into a unified analytical framework and using exploratory data analysis and visualisation techniques to examine these relationships across countries over time. By doing so, it provides a holistic, data-driven perspective on the multifaceted drivers of global life expectancy.

III. Methodology:

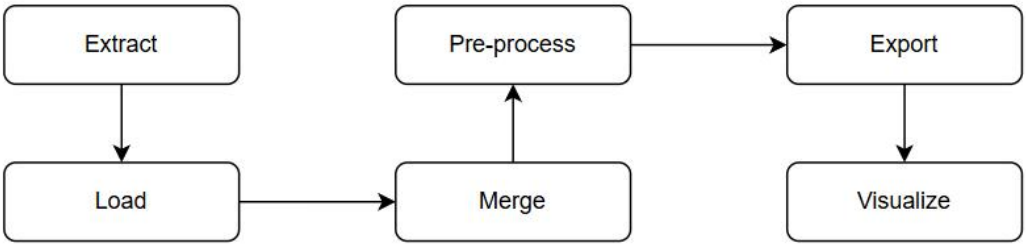
1. Data Sources

The datasets used in this project were obtained from the Our World in Data platform, which aggregates data from reputable international organisations such as the World Health Organization (WHO), World Bank, United Nations (World Population Prospects), and Food and Agriculture Organization (FAO), Global Carbon Budget, UNICEF, Human Mortality Database (HMD). The data were provided in CSV format and included indicators for life expectancy at birth, GDP per capita, healthcare expenditure, vaccination coverage for DTP3, MCV1, and POL3, total carbon emissions, dietary calorie composition, and child mortality rates. The datasets span the period from approximately 2002 to 2022, although coverage varies across countries.

Our World in Data was selected as the primary data source due to its rigorous data curation process and reliance on internationally recognised organisations such as the World Health Organization, World Bank, and United Nations. Using a centralised platform reduced inconsistencies that may arise when sourcing data independently from multiple providers. Additionally, the inclusion of diverse indicators across health, economic, environmental, and nutritional domains enabled a comprehensive

examination of life expectancy, aligning with the multidisciplinary nature of global health research.

2. Data Pipeline Overview



- **Extract & Load**: Raw datasets were first extracted from OWID and loaded into a Jupyter Notebook environment using Python (Pandas) for inspection.
- **Transform**:
  - **Merge**: Datasets were merged using country name, code and year as primary keys.
  - **Preprocess**: Data cleaning and preprocessing steps were applied consistently across indicators.
  - Aggregated regional entities (e.g. “World”, “Africa”, “Europe”) were removed to avoid duplication.
- **Export**: The cleaned and unified dataset was exported to Excel format.
- **Visualise**: Exploratory data analysis and visualisation were conducted using Tableau Desktop.

3. Data Preprocessing

To ensure the data was suitable for analysis, several preprocessing steps were implemented. Rows with missing values were removed entirely, as these gaps could undermine the integrity of the dataset. Removing rows with missing values was chosen over imputation to preserve analytical integrity, as imputing health or economic indicators could introduce artificial patterns. While this approach reduced the number of available observations, it ensured that the resulting dataset reflected only observed values. Column names were updated for clarity, improving the overall accessibility and understanding of the data. In the "Country" column, non-country entities such as continents, regions, and global aggregates were excluded to reduce redundancy and enhance the precision of the analysis, especially in chart visualizations.

Given the varying time ranges across datasets and the differing availability of data for each country, a standardized time range from 2002 to 2022 was established. Countries with fewer than eight years of data (e.g. Tonga, Qatar, Tuvalu, Nauru, Micronesia) were removed to ensure consistent and comparable datasets. Establishing a standardised time range from 2002 to 2022 addressed inconsistencies in data availability across indicators. Countries with insufficient temporal coverage were removed to maintain

```
no_countries = df['Country'].value_counts()
no_countries
```

```
Country
Afghanistan             21
Albania                 21
Algeria                 21
Angola                  21
Antigua and Barbuda     21
                        ..
Micronesia (country)     4
Nauru                    4
Qatar                    4
Tuvalu                   4
Tonga                    3
Name: count, Length: 186, dtype: int64
```

Figure 1 - Number of years with data available per country

comparability although this may bias results towards countries with stronger reporting infrastructures. The use of four-year groupings for vaccination data reduced short-term volatility and enhanced interpretability of long-term trends. Furthermore, vaccination coverage data were grouped into four-year intervals within Tableau to smooth short-term volatility and highlight long-term trends.

4. Visualisation Tools

Tableau Desktop was selected as the primary visualisation tool due to its ability to produce interactive dashboards and geographic maps, facilitating intuitive exploration of global patterns. Its mapping functionality was particularly useful for comparing spatial distributions of life expectancy over time. Python (Pandas) was used for data manipulation, while Tableau was used exclusively for exploratory analysis and presentation-ready visualisations, aligning with the communication-focused objectives of the project.

IV. Result:

1. Global Life Expectancy Trends

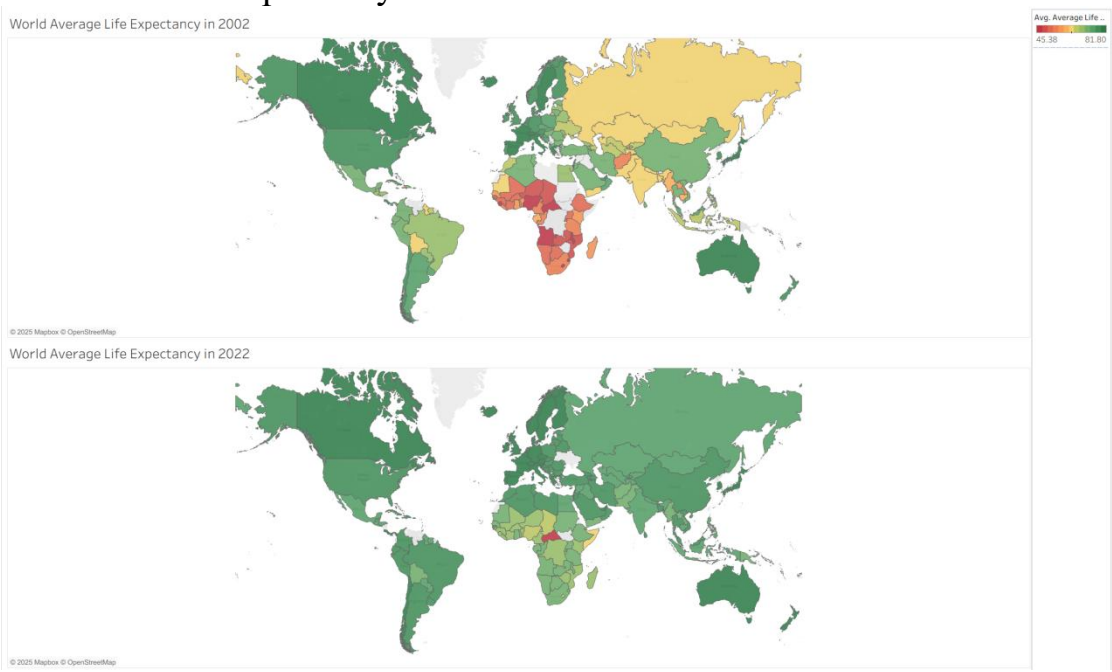


Figure 2 - Average Life Expectancy around the world in 2002 and 2022

World maps comparing average life expectancy in 2002 and 2022 show a widespread increase across most regions over the study period. Substantial improvements are visible in East Asia, South Asia, and most of Africa, while high-income regions maintain consistently higher life expectancy levels. Despite overall progress, notable disparities remain, with several low-income countries continuing to exhibit lower life expectancy compared to high-income nations. These spatial patterns highlight persistent global variation in life expectancy outcomes.

2. GDP, Emissions, and Life Expectancy

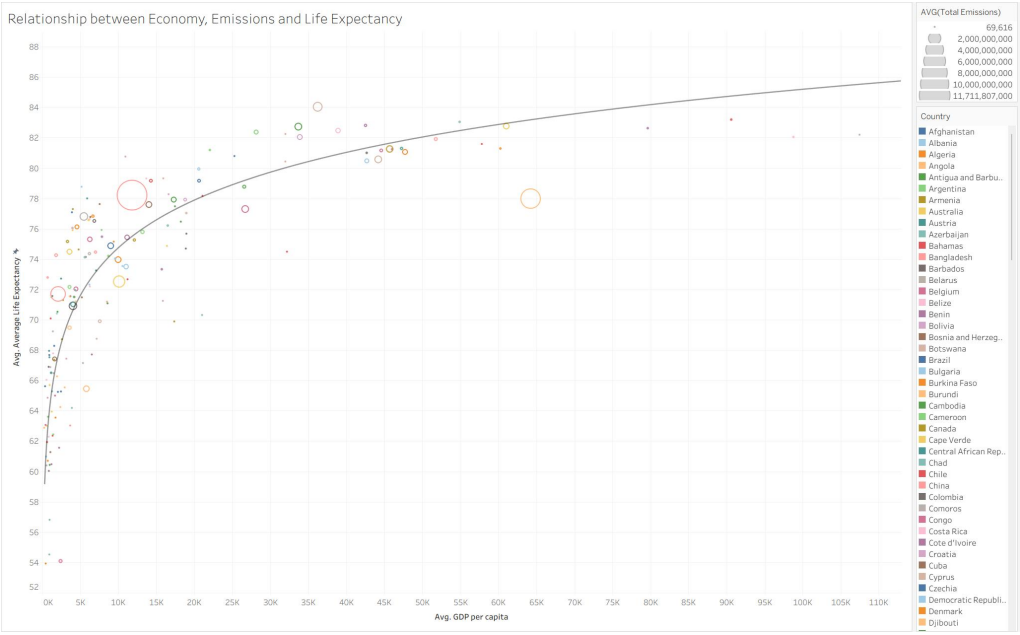


Figure 3 - Relationship between Economy, Emissions and Life Expectancy

Scatter plot analysis demonstrates a strong positive association between GDP per capita and life expectancy, particularly at lower income levels. As GDP increases, life expectancy rises sharply before plateauing at higher income levels, suggesting diminishing returns to health outcomes beyond a certain economic threshold.

A further relationship is observed between GDP per capita and carbon emissions, with higher-income countries generally exhibiting greater emissions. This clustering highlights the link between economic development and environmental impact. While emissions do not show a direct short-term relationship with life expectancy in this analysis, their association with economic activity raises concerns regarding long-term sustainability and population health.

3. Healthcare Spending per Capita, Economic Development, and Life Expectancy

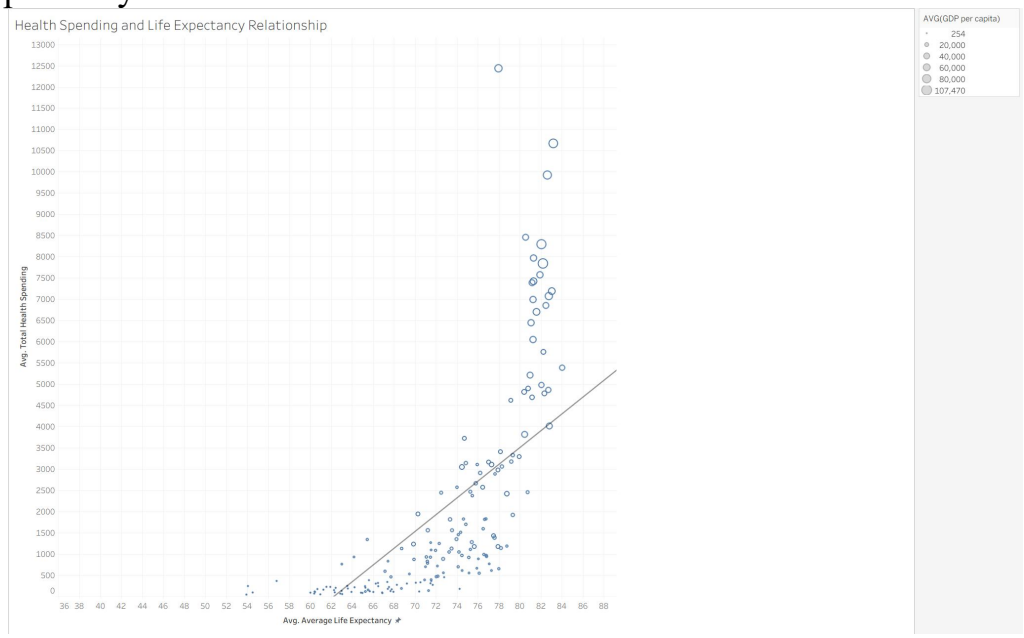


Figure 4 - Relationship between Health Spending and Life Expectancy

Analysis of healthcare expenditure per capita relative to GDP per capita shows that wealthier countries tend to allocate more resources to healthcare. However, when healthcare spending is examined alongside life expectancy, substantial variation is observed among countries with similar spending levels.

Some countries achieve high life expectancy with moderate healthcare expenditure, while others display comparatively lower outcomes despite high spending. This variation suggests that healthcare system efficiency, governance, and preventive care may play a more critical role than expenditure alone in determining health outcomes.

4. Vaccination & Child Mortality

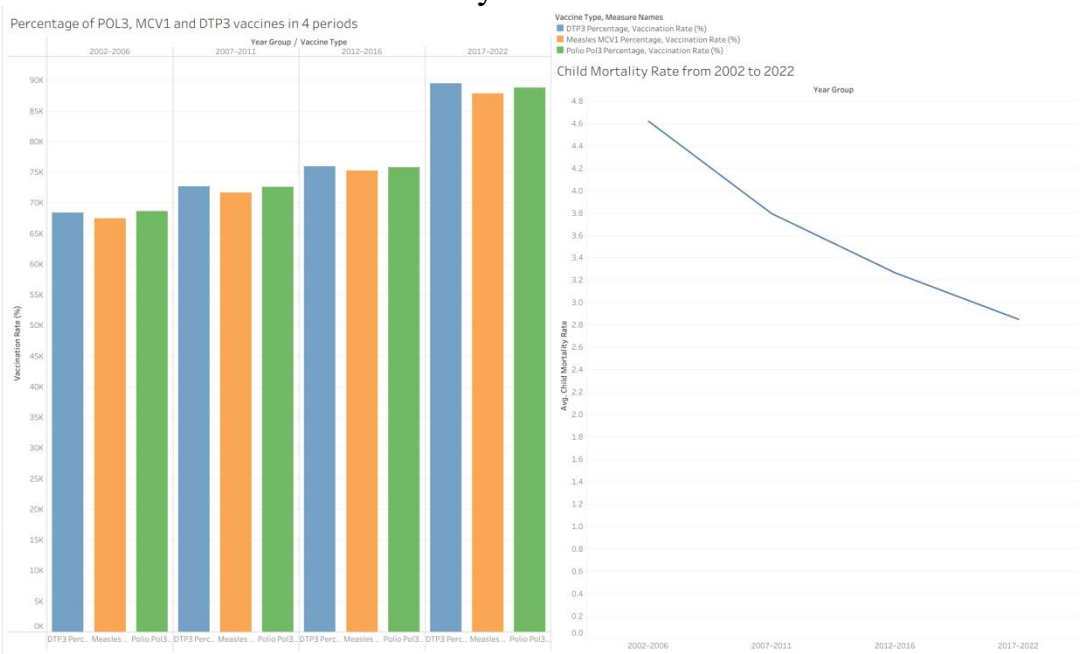


Figure 5 - The inverse of Vaccination Rates and Child Mortality

Bar charts of vaccination coverage for DTP3, MCV1, and POL3, combined with line graphs of child mortality rates, reveal a clear inverse relationship. Over successive time periods, vaccination coverage increases consistently, while child mortality declines.

This pattern is particularly evident in low- and middle-income countries, reinforcing the role of vaccination programmes as one of the most effective public health interventions. The findings support existing evidence that expanding immunisation coverage is strongly associated with improved child survival and, consequently, higher life expectancy.

5. Diet & Calorie Intake

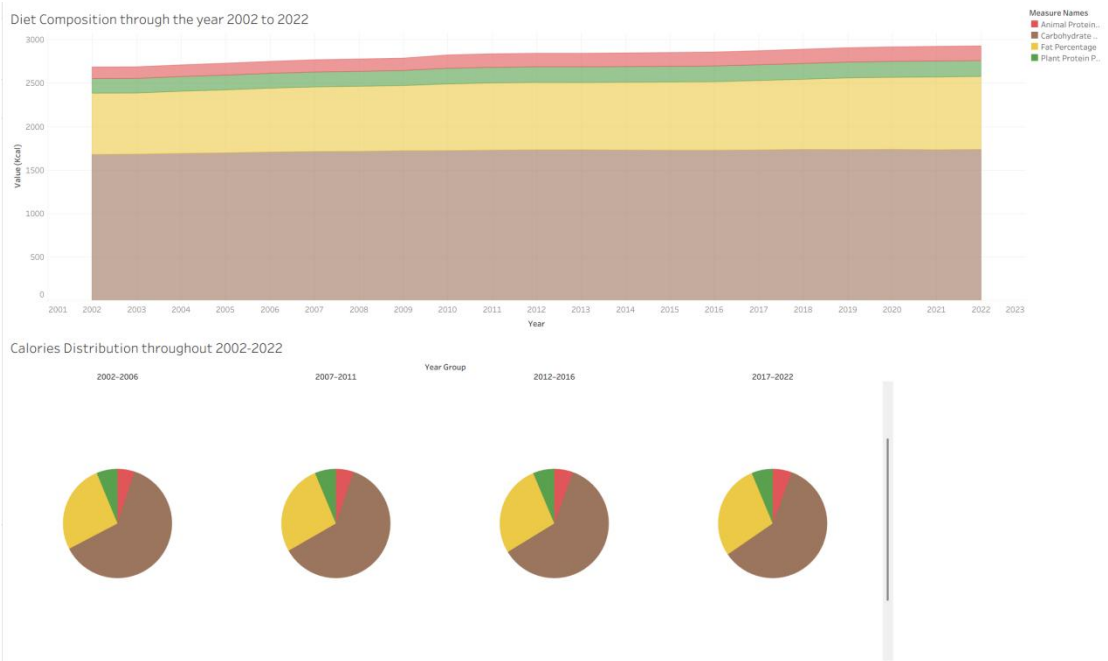


Figure 6 - Total Calorie Consumption and Dietary Compositions throughout the years

Analysis of dietary data indicates that the proportional composition of macro-nutrients remains relatively stable across time periods. In contrast, total calorie intake shows an upward trend. These findings suggest that rising calorie consumption, rather than shifts in dietary composition, may be a more relevant factor in explaining changes in population health within the scope of this aggregated data.

V. Discussion:

1. Interpretation of Findings

The results indicate that economic development remains a strong determinant of life expectancy, particularly in lower-income countries where gains in GDP per capita are associated with substantial improvements in health outcomes. However, the diminishing returns observed at higher income levels suggest that beyond a certain threshold, factors other than income play a more significant role.

Vaccination coverage emerges as one of the most impactful variables, reinforcing its role as a cornerstone of public health policy. The inverse relationship between vaccination rates and child mortality underscores the importance of preventive healthcare interventions. In contrast, the variability observed in healthcare spending per capita suggests that expenditure alone does not guarantee improved outcomes, highlighting differences in healthcare system efficiency and governance structures across countries.

Environmental and dietary indicators exhibit more indirect or limited associations with life expectancy in this analysis. While higher carbon emissions are strongly linked to economic development, their relationship with life expectancy appears less direct in the short term, suggesting that environmental impacts on health may manifest over longer time horizons. Similarly, dietary composition remains relatively stable across countries and time periods, indicating that total consumption levels rather than macro-nutrient distribution may be more relevant in explaining health outcomes. These findings highlight the complexity of health determinants and the limitations of using aggregated global indicators to capture long-term environmental

and nutritional effects.

2. Merits of the Software Application

The software pipeline developed in this project demonstrates the effectiveness of integrating open data sources with analytical and visualisation tools. The modular pipeline allows for easy updates as new data become available and could be extended to include additional indicators. Tableau dashboards provide an intuitive interface for exploring global health trends, making the application suitable for policymakers, educators, and researchers seeking evidence-based insights without requiring advanced technical expertise.

3. Industry / Government Actions

Based on these insights, governments and global health organisations should prioritise expanding vaccination programmes, especially in regions with low coverage. Policymakers should also focus on improving the efficiency of healthcare systems by investing in preventive care and primary health services. Additionally, strategies that promote sustainable economic growth, such as investment in clean energy, are essential to decouple health improvements from environmental degradation.

4. Data Governance Implications

Data Quality:

Data quality represents a significant governance consideration in this project. Although the datasets were sourced from reputable international organisations, missing values and inconsistent reporting across countries and years were evident. These discrepancies reflect differences in national data collection capacity and may bias analyses towards countries with more developed statistical systems. The decision to remove incomplete observations prioritised analytical integrity but reduced coverage, highlighting the trade-off between data completeness and reliability.

Data Lineage and Transparency:

The project involved multiple transformation stages, including merging, filtering, aggregation, and time grouping. Maintaining clear documentation of each step was essential to ensure transparency and reproducibility. Without proper lineage tracking, results could not be independently verified. In real-world applications, such transparency is critical for accountability, particularly when data informs public policy decisions or international development decisions.

Data Security and Integrity:

Although the project uses publicly available, aggregated data, data security and integrity remain relevant governance concerns. Ensuring that datasets are sourced from authoritative platforms, protected from unauthorised modification, and version-controlled is critical to maintaining trust in analytical outputs. In applied industry settings, secure data storage, access controls, and audit logs would be essential to prevent data tampering and ensure analytical accountability.

Ethics and Responsible Use:

Although the dataset contains only aggregated, non-personal data, ethical considerations remain relevant. Correlation-based findings risk being misinterpreted

as causal, potentially leading to inappropriate policy conclusions. Care was therefore taken to present results descriptively and acknowledge limitations. Additionally, global comparisons must be communicated sensitively to avoid stigmatising countries with lower life expectancy.

VI. Conclusions:

This project set out to explore the global determinants of life expectancy through the integration and analysis of multiple open datasets. By developing a complete data pipeline and applying exploratory visualisation techniques, the study identified clear relationships between life expectancy and economic development, vaccination coverage, healthcare spending, and environmental factors.

The findings highlight the continued importance of preventive healthcare interventions, particularly vaccination programmes, and emphasise the need for efficient allocation of healthcare resources rather than increased expenditure alone. At the same time, the association between economic growth and increased emissions underscores the necessity of sustainable development strategies in improving long-term population health.

Several limitations must be acknowledged. The analysis relies on country-level averages, which may mask sub-national inequalities, and is constrained by missing data and inconsistent reporting across countries. In addition, the study is based on correlation rather than causal modelling, limiting the ability to infer direct cause–effect relationships.

Future work could address these limitations by incorporating sub-national health indicators, more granular dietary and environmental data, or applying machine learning and causal inference techniques to predict life expectancy. Overall, this project demonstrates the value of data-driven analysis in supporting evidence-based decision-making for global health policy.

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